

1 Problem Formulation

Retrieval-augmented generation (RAG) systems commonly rely on text extracted from documents to retrieve relevant context. This study compares a text-only RAG baseline with an enhanced multimodal architecture that integrates textual, visual, and knowledge-graph information. The objective is to understand how the architectures perform across different question-answering scenarios and determine whether multimodal and structured knowledge improves retrieval and answer quality.

2 Data Preprocessing Technique

A 100-page subset of the `small-publaynet-wds` dataset was selected as a proof-of-concept corpus. This provided sufficient content for comparing the architectures while keeping preprocessing, indexing, knowledge-graph construction, API token usage, and associated costs manageable.

- **Text extraction:** Optical character recognition (OCR) was required because PubLayNet provides page images and layout annotations rather than machine-readable PDF text.
- **Fixed-size chunking:** Since the text was recovered through OCR rather than extracted from a native digital document, it may contain incorrect characters, missing punctuation, merged words, and unreliable sentence boundaries. Fixed-size overlapping chunks were therefore used as a simple and consistent method that does not depend on clean paragraph or sentence structure. Each chunk retained its source-page identifier for provenance.
- **Multimodal embeddings:** Text chunks were embedded using `text-embedding-3-small`, which provides a suitable balance between semantic retrieval quality and computational cost for the proof-of-concept corpus. Figures and tables were embedded using `CLIP ViT-B/32`, which places text and images in a shared embedding space and therefore enables natural-language queries to retrieve relevant visual content.
- **Vector indexing:** Text embeddings were indexed using FAISS with cosine similarity. FAISS was chosen as a lightweight and efficient alternative to deploying a larger vector database for only 100 pages.

3 Model Architecture

All architectures use `gpt-4o` for answer generation, ensuring that performance differences reflect the supplied evidence rather than model capability. Evaluation is performed separately using `gpt-4.1`.

3.1 Baseline

The baseline is a text-only RAG pipeline. The query is embedded, the top- k ($k = 3$) chunks are retrieved from a FAISS index, and `gpt-4o` generates an answer using only the retrieved text, abstaining when the answer is unavailable.

3.2 Enhanced

The enhanced architecture adds two components through late fusion:

- **Knowledge graph.** An LLM extracts cached subject–relation–object triples, which are stored in a NetworkX graph. At query time, matched entities and neighbouring triples are added to the generation prompt as structured facts.
- **Multimodal retrieval.** CLIP retrieves the most relevant figure or table. The caption-mediated variant supplies its caption to `gpt-4o`, while the VLM variant supplies the image itself alongside the retrieved text for direct visual reasoning.

3.3 Explainability:

Each response identifies the supporting evidence used to generate the answer, including the relevant text chunk, source document and page, and, where applicable, the retrieved figure or table. It also provides a brief explanation of how the retrieved evidence was combined to answer the question.

4 Experimental Evaluation

Three question sets were constructed: *text questions*, answerable from a single passage; *figure questions*, requiring information from a figure or table; and *multi-hop questions*, requiring evidence from multiple passages or sections. For each category, **gpt-4o** generated questions from the selected documents using three to five few-shot examples. Each item contained a question, reference answer, expected source, and supporting evidence. All items were manually reviewed to verify the ground truth and ensure that the wording did not reveal the required modality.

Retrieval is evaluated using Recall@ k , which checks whether the expected source appears in the top- k results, and Mean Reciprocal Rank (MRR), which rewards higher-ranked relevant sources. Generated answers are assessed for accuracy, faithfulness, and relevancy using **gpt-4.1**, a different model from the generator to reduce self-evaluation bias. Faithfulness measures whether the answer is supported by the retrieved evidence, while relevancy measures whether it directly addresses the question.¹

Question generation uses a fixed seed and temperature 0. The implementation, evaluation scripts, question sets, and reproduction instructions are available in the accompanying GitHub repository.²

5 Results and Discussion

Table 1: Four-way ablation by question type. Retrieval (Recall@3 / MRR) and generation (Accuracy / Faithfulness / Relevancy). Best accuracy per type in bold.

Question type	System	R@3	MRR	Acc.	Faith.	Rel.
Text Only	baseline	0.958	0.938	0.833	0.833	1.000
	+KG	0.958	0.938	0.833	0.833	1.000
	+multimodal	0.958	0.938	0.833	0.875	1.000
	+both	0.958	0.938	0.833	0.875	1.000
Image	baseline	0.000	0.000	0.086	0.314	0.457
	+KG	0.000	0.000	0.057	0.257	0.343
	+multimodal	0.486	0.400	0.514	0.343	0.657
	+both	0.486	0.400	0.457	0.400	0.629
Multi-hop	baseline	1.000	1.000	0.500	0.706	0.941
	+KG	1.000	1.000	0.559	0.706	0.941
	+multimodal	1.000	1.000	0.588	0.853	0.941
	+both	1.000	1.000	0.618	0.794	0.971

Text Only. All systems perform identically (accuracy 0.833), confirming the enhancements neither help nor harm when the answer is already in the retrieved text. This validates the comparison: gains elsewhere are attributable to the question type, not to the enhancements being generally stronger.

Image (Figures and Tables). The multimodal systems lift accuracy from ≈ 0.09 to 0.51 (a roughly six-fold gain), while text-only systems score near zero, as they cannot retrieve images (Recall = 0 by construction). Accuracy closely tracks image retrieval (0.486), showing CLIP retrieval is the principal bottleneck: on dense, visually similar scientific figures it returns the correct figure only about half the time, capping downstream accuracy. The KG slightly *degrades* figure performance (0.514 to 0.457 for +both), as its text-derived facts are irrelevant to visual queries and mildly distract the generator.

Multi-hop. These questions combine two facts from different parts of the *same* page, so retrieving that one page satisfies them; this is why retrieval is perfect (Recall = 1.0) and matches the text-only case, with the difficulty lying in generation rather than retrieval. Accuracy rises steadily (0.500 baseline to 0.618 for +both), the KG contributing by supplying connected facts. Its benefit is thus conditional on retrieval sufficiency: when the needed text is already retrieved, structured facts help complete multi-fact answers; on single-fact questions they are redundant.

Overall. Each enhancement helps the question type it targets: multimodal is essential for Image-resident answers, the KG aids multi-fact reasoning. The combined system is best on text and multi-hop, while multimodal-only is best on figures (where the KG distracts). The main limitations are CLIP retrieval on dense scientific figures, which caps visual accuracy, and OCR noise, which degrades text and KG-entity quality.

¹S. Es, J. James, L. E. Anke, and S. Schockaert, “Ragas: Automated Evaluation of Retrieval Augmented Generation,” in *Proc. EACL System Demonstrations*, 2024, pp. 150–158.

²<https://github.com/sokistar24/multimodal-graph-rag>